

EXPECTED VALUES (2D RANDOM VARIABLES)

$$1D: E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$E(g(x)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

$$E(g(x, y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) \underbrace{f_{XY}(x, y)}_{\text{JOINT PDF}} dx dy$$

↑
SOME FUNCTION
OF X AND Y.

JOINT PDF

$$E(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{xy}(x, y) dx dy$$


$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{xy}(x, y) dy dx$$

$$= \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{xy}(x, y) dy \right] dx$$

MARGINAL

$$= \int_{-\infty}^{\infty} x f_x(x) dx$$

IF $E(XY) = E(X)E(Y)$, THE RANDOM
VARIABLES ARE UNCORRELATED.

INDEPENDENT $\Rightarrow$ UNCORRELATED

THE REVERSE IS NOT TRUE!

IF $E(XY) \neq E(X)E(Y)$ THE R.V.S.
ARE CORRELATED.

IF $E(X)E(Y) = E(X)E$

$$E(\overset{\text{CONSTANT}}{\downarrow} aX + \overset{\text{CONSTANT}}{\downarrow} bY) = aE(X) + bE(Y)$$

WE GIVE A SPECIAL NAME TO

$E(XY)$: THE CORRELATION OF X AND Y .

$$= \iint xy f_{XY}(x,y) dx dy$$

IN THE SPECIAL CASE THAT $f_{XY}(x,y) = f_X(x) f_Y(y)$ INDEPENDENT
↓

$$E(XY) = \iint xy f_X(x) f_Y(y) dx dy = \int x f_X(x) dx \int y f_Y(y) dy$$

$\downarrow$
 $E(X)$

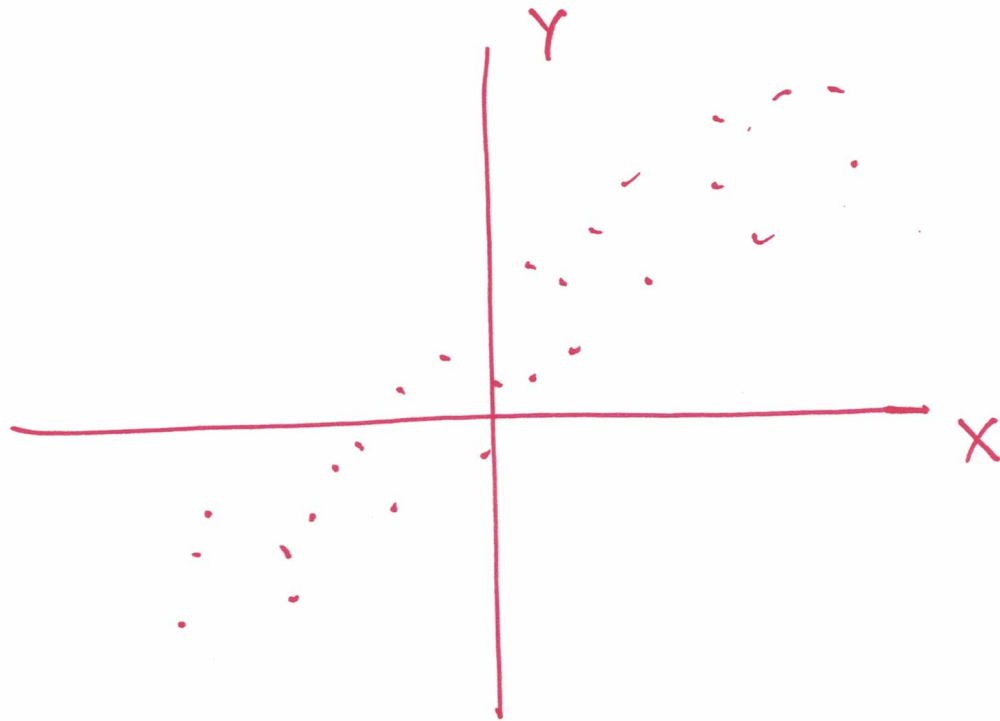
$\downarrow$
 $E(Y)$

i.e. THE RV ARE UNCORRELATED.

IN CORRELATED IS WEAKER THAN INDEPENDENT.

CORRELATED $\Rightarrow$ THERE'S A LINEAR

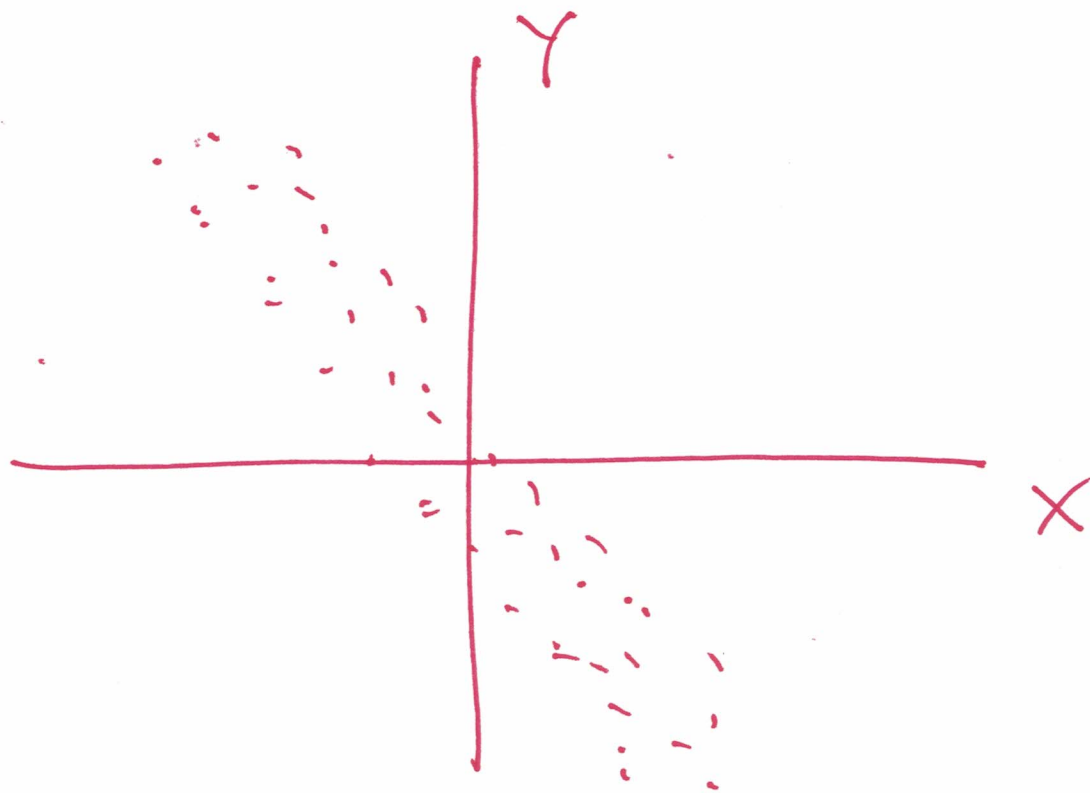
RELATIONSHIP BETWEEN X AND Y



X BIGGER THAN
ITS MEAN

$\Rightarrow$ Y IS
LIKELY BIGGER
THAN ITS
MEAN

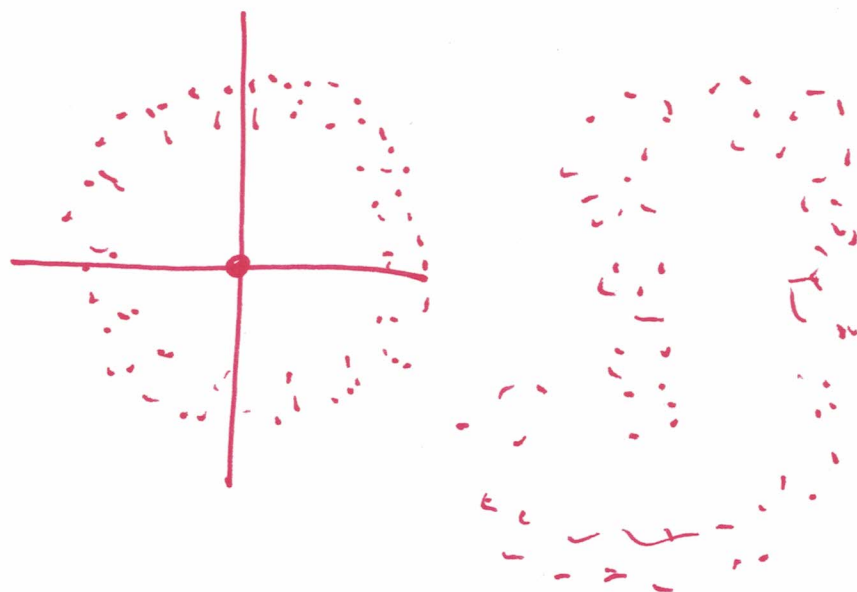
POSITIVE
CORRELATION



IF X IS $>$ ITS
MEAN, Y IS
LIKELY TO BE
LESS THAN ITS
MEAN

NEGATIVELY
CORRELATED.

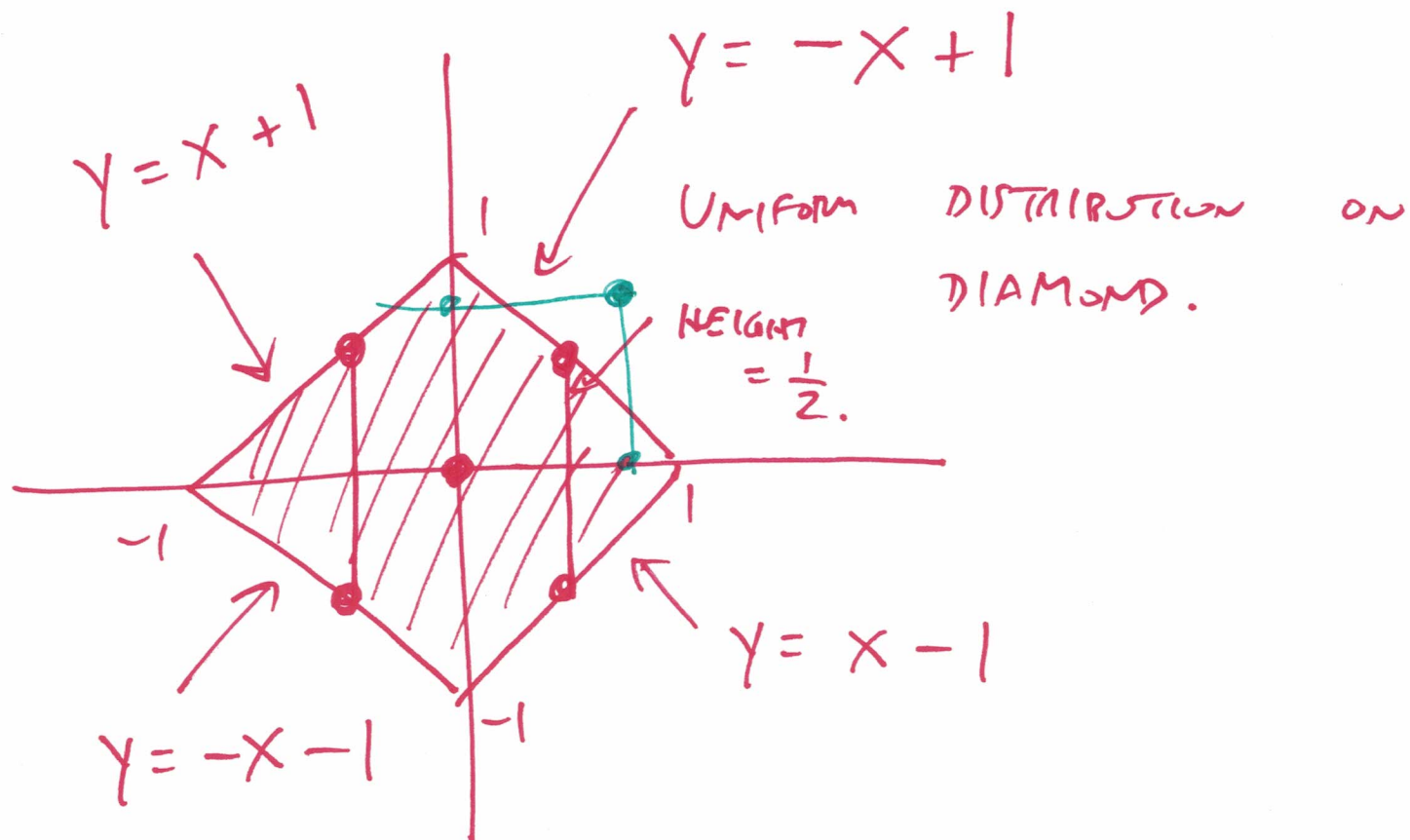
TWO RVs CAN
STILL BE RELATED
BUT NOT CORRELATED



INDEPENDENCE $\Rightarrow$ THERE IS NO
CONNECTION BETWEEN THE R.VS.
AT ALL, LINEAR OR OTHERWISE.

$$Y = 2X$$

$$Y = |X|$$



$$E(XY) = \iiint_{\text{DIAMOND}} \frac{1}{2} xy \, dx \, dy$$

$$= \int_{-1}^0 \int_{-x-1}^{x+1} \frac{1}{2} xy \, dy \, dx + \int_0^1 \int_{x-1}^{1-x+1} \frac{1}{2} xy \, dy \, dx$$

$$\int_0^1 \int_{x-1}^{1-x} \frac{1}{2} xy \, dy \, dx = \int_0^1 \left[\frac{1}{4} x y^2 \right]_{y=x-1}^{y=1-x} dx$$

$$= \int_0^1 \frac{1}{4} x [(1-x)^2 - (x-1)^2] dx$$

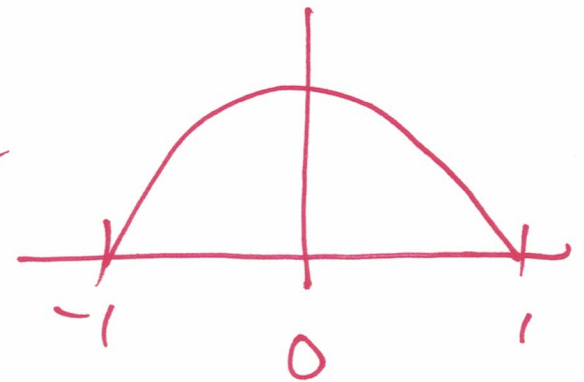
ϕ .

$$E(XY) = \phi.$$

MARGINAL IN X

$$E(X) = \phi.$$

$$E(Y) = \phi$$



UNCORRELATED

BUT X AND Y ARE NOT
INDEPENDENT.

WE WOULD NEED

$$f_{XY}(x,y) = f_X(x) f_Y(y)$$

$$f_{XY}\left(\frac{3}{4}, \frac{3}{4}\right) \stackrel{?}{=} f_X\left(\frac{3}{4}\right) f_Y\left(\frac{3}{4}\right)$$

↓

$\emptyset$

$\neq$

||

NOT $\emptyset$. NOT $\emptyset$

NOT IN

DIAMOND

ANOTHER KEY TERM:

THE COVARIANCE OF X AND Y

$$\text{IS } E((X - \mu_X)(Y - \mu_Y))$$

$$= E(XY - \mu_X Y - X\mu_Y + \mu_X \mu_Y)$$

$$= E(XY) - \underbrace{\mu_X E(Y)}_{\mu_Y} - \underbrace{\mu_Y E(X)}_{\mu_X} + \mu_X \mu_Y$$

$$= E(XY) - E(X)E(Y)$$

IF X AND Y ARE UNCORRELATED, $\Leftrightarrow$
THE COVARIANCE IS $\emptyset$

IF $Y = X$

$$E(X^2) - (E(X))^2$$

THE VARIANCE OF X

• GIVEN $f_{XY}(x, y)$

ARE X AND Y UNCORRELATED?

ARE X AND Y INDEPENDENT?

WHAT IS THE CORRELATION COEFFICIENT?

THE CORRELATION COEFFICIENT OF
X AND Y IS

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\sigma_X = \sqrt{\text{Var}(X)}$$

$$\sigma_Y = \sqrt{\text{Var}(Y)}$$

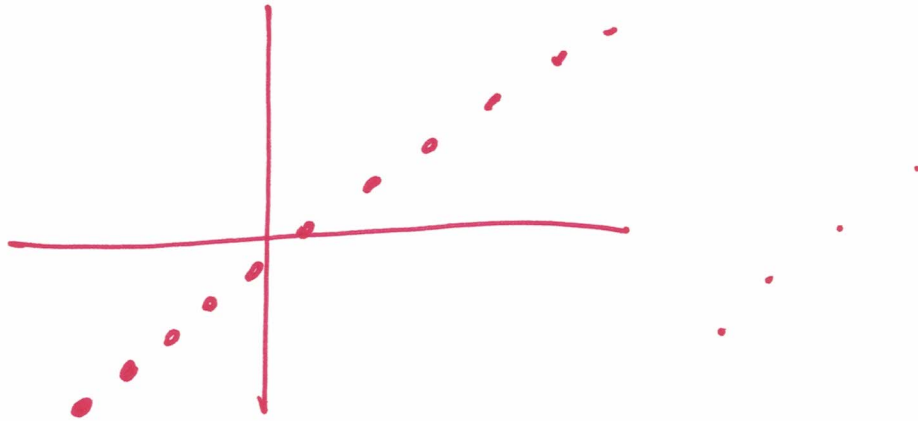
↑
"rho"

WHILE THE VALUE OF $\text{Cov}(X, Y)$
COULD BE ANYTHING,

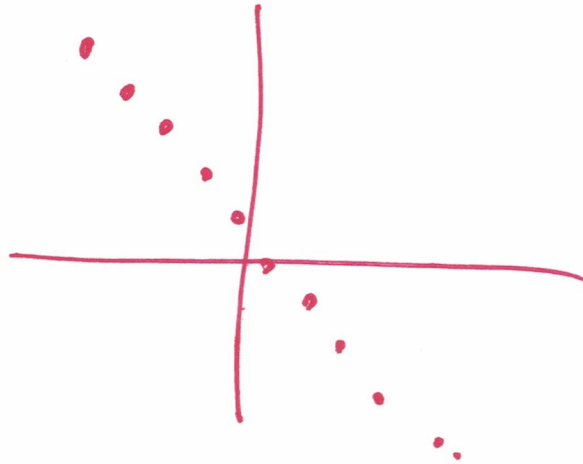
ρ_{XY} IS LIMITED TO THE RANGE
 $[-1, +1]$

$$\rho_{xy} = +1$$

$$Y = (\text{POSITIVE CONSTANT}) X + b$$



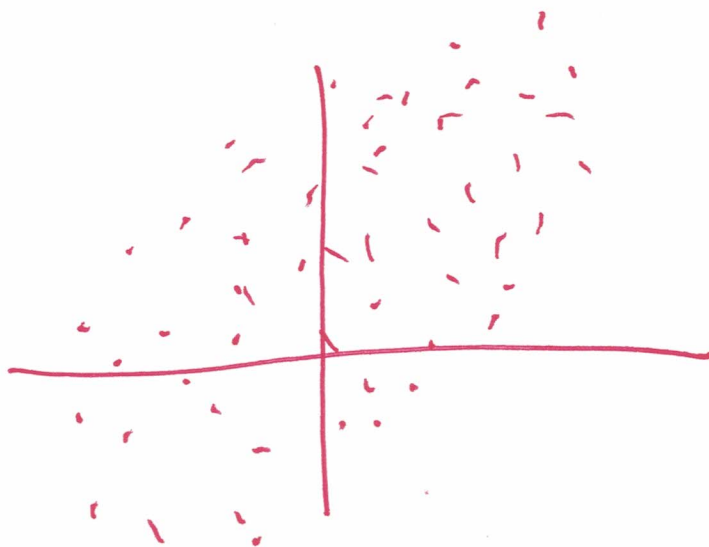
$$\rho_{xy} = -1$$



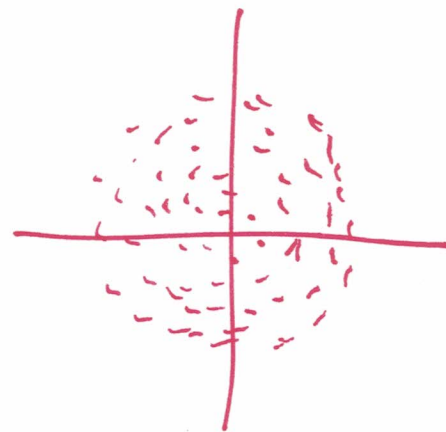
$$\rho_{xy} = 0.8$$



$$\rho_{xy} = 0.2$$



$$\rho_{xy} = \phi$$



THIS IS THE SAME ρ AND THE SAME
IDEA FROM LAST TIME'S JOINT GAUSSIAN.

$$f_{xy}(x,y) = \frac{1}{2\pi|\Sigma|} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

(*) ONLY FOR THE JOINT
GAUSSIAN PDF

UNCORRELATED $\Rightarrow$
 $\rho=0$

INDEPENDENT

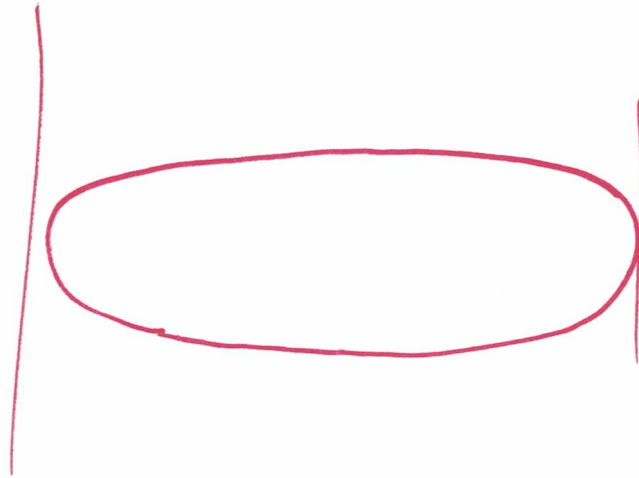
$$f_{xy} = f_x f_y$$

$$\Sigma = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix}$$

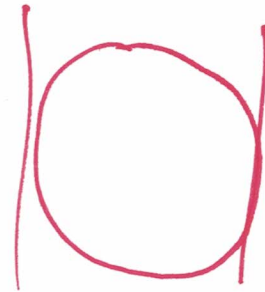
THE CORRELATION
COEFFICIENT.



A



B



C

INCREASING

ρ :

C

B

A

INCREASING

σ_x :

A

C

B